

```
# Install required libraries
!pip install -q torch transformers datasets peft accelerate

# Import libraries
import torch
from transformers import AutoModelForSequenceClassification, AutoTokenizer
from datasets import load_dataset
from peft import LoraConfig, get_peft_model
```

```
model_name = "distilbert-base-uncased"

# Load tokenizer
tokenizer = AutoTokenizer.from_pretrained(model_name)

# Load model
model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2
)

print("Model loaded successfully!")
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable faster and secure connections.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub.
config.json: 100% 483/483 [00:00<00:00, 11.2kB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 898B/s]

vocab.txt: 232k/? [00:00<00:00, 3.80MB/s]

tokenizer.json: 466k/? [00:00<00:00, 9.07MB/s]

model.safetensors: 100% 268M/268M [00:02<00:00, 214MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 264.12it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased

Key	Status
-----+-----	-----+-----
vocab_transform.bias	UNEXPECTED
vocab_layer_norm.bias	UNEXPECTED
vocab_layer_norm.weight	UNEXPECTED
vocab_projector.bias	UNEXPECTED
vocab_transform.weight	UNEXPECTED
classifier.bias	MISSING
pre_classifier.bias	MISSING
classifier.weight	MISSING
pre_classifier.weight	MISSING

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expected it to be there.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training from scratch.

Model loaded successfully!

```
# Load dataset (IMDB sentiment dataset)
dataset = load_dataset("imdb")

# Tokenization function
def tokenize_function(example):
    return tokenizer(example["text"], truncation=True, padding="max_length")
```

```
# Apply tokenization
tokenized_dataset = dataset.map(tokenize_function, batched=True)

# Reduce dataset size for quick training
train_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(500))
test_dataset = tokenized_dataset["test"].shuffle(seed=42).select(range(200))

print("Dataset ready!")
```

```
README.md:      7.81k/? [00:00<00:00, 281kB/s]

plain_text/train-00000-of-      21.0M/21.0M [00:00<00:00, 105MB/s]
00001.parquet: 100%

plain_text/test-00000-of-      20.5M/20.5M [00:00<00:00, 102MB/s]
00001.parquet: 100%

plain_text/unsupervised-00000-of-      42.0M/42.0M [00:00<00:00, 210MB/s]
00001.p(...): 100%

Generating train split: 100%      25000/25000 [00:00<00:00, 61132.55 examples/s]
Generating test split: 100%      25000/25000 [00:00<00:00, 70000.04 examples/s]
Generating unsupervised split: 100%      50000/50000 [00:00<00:00, 84475.35 examples/s]

Map: 100%      25000/25000 [00:31<00:00, 749.50 examples/s]
Map: 100%      25000/25000 [00:30<00:00, 810.37 examples/s]
Map: 100%      50000/50000 [01:10<00:00, 851.28 examples/s]

Dataset ready!
```

```
lora_config = LoraConfig(
    r=8,
    lora_alpha=16,
    target_modules=["q_lin", "v_lin"],
    lora_dropout=0.1,
    bias="none",
    task_type="SEQ_CLS"
)

print("LoRA Config created!")
```

LoRA Config created!

```
peft_model = get_peft_model(model, lora_config)
```

```
# Show trainable parameters
peft_model.print_trainable_parameters()
```

WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernel trainable params: 739,586 || all params: 67,694,596 || trainable%: 1.0925

```
from transformers import TrainingArguments, Trainer
```

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)
```

```
trainer = Trainer(
```

```

        model=peft_model,
        args=training_args,
        train_dataset=train_dataset
    )

    trainer.train()

```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' is deprecated. Please use 'pin_memory_device' instead.
 super().__init__(loader)

[63/63 20:31, Epoch 1/1]

Step	Training Loss
10	0.710151
20	0.669138
30	0.679355
40	0.642490
50	0.653073
60	0.640813

TrainOutput(global_step=63, training_loss=0.6637767080276732, metrics={'train_runtime': 1264.2133, 'train_samples_per_second': 0.396, 'train_steps_per_second': 0.05, 'total_flos': 67369703424000.0, 'train_loss': 0.6637767080276732, 'epoch': 1.0})

```

# Evaluate model
eval_results = trainer.evaluate(eval_dataset=test_dataset)
print(eval_results)

# Test prediction
text = "The movie was fantastic and inspiring!"
inputs = tokenizer(text, return_tensors="pt")

outputs = peft_model(**inputs)
prediction = torch.argmax(outputs.logits, dim=1)

print("Predicted label:", prediction.item())

```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' is deprecated. Please use 'pin_memory_device' instead.
 super().__init__(loader)

[25/25 02:32]

{'eval_loss': 0.6324213147163391, 'eval_runtime': 160.2948, 'eval_samples_per_second': 1.248, 'eval_loss_std': 0.0001, 'epoch': 1.0}
 Predicted label: 1

